



Research Paper

Oblique-fin heat sink design for uniform CPU cooling in immersion-cooled servers

M. Khoshvaght-Aliabadi^{a,b,c,*}, A. Nasrolahzadeh^d, P. Ghodrati^a, Y.T. Kang^{a,c,**}^a School of Mechanical Engineering, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea^b Department of Chemical Engineering, Sha.C., Islamic Azad University, Shahrrood, Iran^c Research Center for Plus Energy Building Innovative Technology, School of Mechanical Engineering, Korea University, 145 Anam-ro, Seongbuk-Gu, Seoul 02841, South Korea^d Department of Chemical Engineering, Faculty of Engineering, Ferdowsi University of Mashhad, Iran

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ABSTRACT

This study investigates single-phase immersion cooling (SPIC) as an advanced thermal management strategy for multi-CPU server platforms, focusing on the Intel® Server Board S2600BPB equipped with oblique-fin heat sinks arranged in novel configurations. Using FC-40 as the coolant, the research explores the combined influence of heat sink design and coolant flow rates on CPU temperature, temperature uniformity index (TUI), and pumping power across a broad range of CPU thermal design powers (TDPs). The proposed oblique-fin designs, featuring converging channel formations, demonstrate notable improvements in temperature uniformity and cooling efficiency, with the most significant enhancement observed at lower flow rates. Under a 500 W TDP, the design achieves CPU temperature reductions of up to 7 °C and TUI improvements of up to 20 %. This configuration achieves a favorable trade-off between thermal performance and hydraulic efficiency, incurring less than a 3 % increase in pumping power despite enhanced heat removal capabilities. Moreover, the design effectively dissipates heat from CPUs and key downstream elements, including the Platform Controller Hub (PCH) and Voltage Regulator (VR), alleviating thermal cascade effects typical in server systems. Increasing the coolant flow rate from 2 to 7 L/min at 500 W TDP results in temperature reductions of approximately 8.7 °C for DIMMs, and more substantial decreases of 18.5 °C and 16.4 °C for the PCH and VR, respectively. This work establishes oblique-fin heat sinks as a promising solution for uniform, efficient thermal management in immersion-cooled servers.

1. Introduction

The exponential growth of emerging technologies including big data, artificial intelligence, and 5G has positioned data centers (DCs) as essential infrastructure for supporting the digital economy [1]. Since the development of the first integrated circuit in 1958 [2], the number of transistors integrated into increasingly smaller chips has grown dramatically. While this miniaturization has significantly enhanced server performance, it has also resulted in a substantial increase in heat flux, posing major cooling challenges. Modern high-performance chips can experience heat fluxes of up to 1 kW/cm² [3], with thermal design power (TDP) exceeding 1000 W [4]. Concurrently, the rapid and often unpredictable evolution of information technology has driven a considerable rise in data storage and processing demands, leading to a

continuous escalation in global energy consumption. For instance, between 1995 and 2015, worldwide energy usage increased markedly from 8588.9 to 13,147.3 million tons of oil equivalent (Mtoe) [5].

Reflecting this trend, the number of operational high-performance hyperscale DCs has increased rapidly, reaching 700 by 2021 [6]. Consequently, DCs' energy demand is anticipated to grow sharply from 286 TWh in 2016 to 752 TWh in 2030 mainly owing to the substantial electricity required for cooling. [7]. It has been reported that approximately 35–40 % of the total energy demand in electronic systems is devoted to cooling operations, underscoring a considerable potential for energy savings if cooling power is effectively managed [8]. Elevated temperatures not only reduce the lifespan of electronic components but also increase the likelihood of malfunctions. Studies suggest that nearly 55 % of electronic failures can be attributed to excessive heat [9]. Therefore, the implementation of efficient cooling solutions has become

* Corresponding author at: School of Mechanical Engineering, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea.

** Corresponding author at: The School of Mechanical Engineering, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea.

E-mail addresses: mkhaliabadi@gmail.com, mkhaliabadi@korea.ac.kr (M. Khoshvaght-Aliabadi), ytkang@korea.ac.kr (Y.T. Kang).

Nomenclature		<i>max</i>	maximum
		<i>min</i>	minimum
		<i>str</i>	straight
c_p	specific heat capacity (J/kg·°C)		
g	gravitational acceleration (m ² /s)		
p	pressure (Pa)		
P_P	pumping power (W)		
S	volumetric heat source (W/m ³)		
T	temperature (°C)		
u	velocity (m/s)		
V	volumetric flow rate (m ³ /s)		
<i>Greek symbols</i>			
μ	dynamic viscosity (Pa·s)		
ρ	density (kg/m ³)		
λ	thermal conductivity (W/m·°C)		
<i>Subscripts</i>			
<i>con</i>	converging		
<i>div</i>	diverging		
<i>eff</i>	effective		
<i>f</i>	fluid		
		<i>Acronyms</i>	
		A	Arrangement
		AI	Artificial Intelligence
		CRPS	Common Redundant Power Supply
		CPU	Central Processing Units
		DC	Data Center
		DIMM	Dual Inline Memory Module
		HPC	High-Performance Computing
		HS	Heat Sink
		PCB	Printed Circuit Board
		PCH	Platform Controller Hub
		SPIC	Single-Phase Immersion Cooling
		TDP	Thermal Design Power (W)
		TUI	Temperature Uniformity Index
		VR	Voltage Regulator

essential for ensuring the reliability, longevity, and energy efficiency of modern DCs.

Addressing the substantial power demands of DCs, particularly those associated with cooling, has driven the adoption of diverse thermal management strategies, including forced air cooling [10], water-cooling plates [11], and immersion cooling [12]. Air cooling, the most widely employed method, uses fans to circulate air as the primary coolant over electronic components [13]. However, with increasing server power density, this conventional approach is becoming inadequate, facing challenges such as elevated temperatures, high humidity, and excessive energy consumption [14]. Indirect liquid cooling using water-cooling plates can enhance heat transfer and improve energy efficiency, but such systems are typically heavy and occupy considerable space. In contrast, immersion cooling offers superior thermal performance [12] and supports higher server densities due to its compact design and efficient heat removal capabilities [15]. Consequently, immersion cooling is expected to become the predominant approach for liquid cooling in future DC applications.

Immersion cooling technologies are broadly categorized into two categories: single-phase immersion cooling (SPIC) [16,17] and two-phase immersion cooling (TPIC) [18,19]. Experimental studies by Kanbur et al. [20] demonstrated that TPIC systems outperform SPIC systems by 72–79 % in terms of performance trend coefficients. Nevertheless, the greater complexity, maintenance requirements, and higher costs of TPIC systems limit their large-scale implementation, making SPIC systems a more practical choice for widespread deployment [15,21]. Hnayno et al. [22] reported that, for high-heat-flux servers, SPIC technology enhances energy efficiency, reducing per-server energy demand by at least 20 % compared with air-cooled systems. Cheng et al. [17] designed a pump-driven SPIC system with a finned heat sink to investigate heat transfer characteristics under varying coolant flow characteristics and heat sink materials. Their findings demonstrated that increasing the flow rate from 0.4 to 1.2 m/s reduced central processing unit (CPU) temperature by 16.1 %, indicating a negative correlation between coolant velocity and CPU temperature. Muneeshwaran et al. [23] conducted an optimization study on CPU heat sinks in SPIC systems and showed that replacing conventional designs with heat pipes can reduce case temperature by approximately 3.3 °C under specified operating conditions. Similarly, Shrigondekar et al. [24] experimentally analyzed different inlet and outlet configurations, as well as fin spacing, in a SPIC system. Their results revealed that a T-type configuration using FC-40 coolant

outperformed the Z-type design, achieving a 12.6 % reduction in thermal resistance. Khoshvaght-Aliabadi et al. [25] optimized heat sink designs and coolant parameters for SPIC-configured server CPUs, showing that proper geometric and operational adjustments can sustain CPU temperatures below critical limits even under high TDPs. They also examined various coolant distribution strategies for multi-CPU servers and showed that optimized cold plate configurations can significantly improve temperature uniformity while reducing pumping power requirements [26]. Saini et al. [27] focused on optimizing chip heat sink structures to minimize thermal resistance and pressure drop. Furthermore, Chen et al. [28] investigated CPU heat dissipation using heat sinks and found that a novel three-dimensional horizontal fin arrangement outperformed conventional configurations in terms of thermal performance.

Despite extensive research on SPIC, existing studies have predominantly focused on operating conditions [29], coolant selection [30], and conventional straight-fin heat sink geometries [31]. In such designs, the flow channels maintain an approximately uniform cross-section along the coolant path, resulting in nearly uniform coolant velocity and a spatially invariant convective heat transfer coefficient. While this approach is effective at moderate thermal loads, it becomes increasingly limited for multi-CPU server architectures operating at elevated TDPs. As the coolant progressively absorbs heat while flowing downstream, the local temperature driving potential diminishes, often leading to thermal accumulation, non-uniform CPU temperatures, and degraded cooling performance of downstream components. These challenges are further exacerbated in immersion environments employing viscous dielectric fluids, where increasing the bulk flow rate to compensate for downstream thermal degradation is constrained by pumping power considerations.

Consequently, there remains a critical research gap in understanding how heat sink geometry can be deliberately leveraged to modulate local fluid dynamics and mitigate these limitations in immersion-cooled servers. Most prior studies have concentrated on conventional heat sink designs, low-to-medium TDP processors, or isolated CPU cooling, with limited attention given to multi-component thermal interactions involving CPUs, dual in-line memory modules (DIMMs), platform controller hubs (PCHs), and voltage regulators (VRs). In particular, the potential of oblique-fin heat sinks to improve temperature uniformity and downstream cooling effectiveness through controlled internal flow redistribution has not been systematically explored.

Motivated by this physical and practical limitation, the present study adopts oblique-fin heat sinks as a targeted flow-modulation strategy. By angling the fins relative to the main flow direction, converging and diverging internal channels are formed, enabling controlled acceleration or deceleration of the coolant along the flow path. Through comprehensive numerical simulations, this work systematically evaluates the thermal and hydraulic performance of these configurations across a wide range of CPU TDPs, with emphasis on temperature uniformity, multi-component thermal interactions, and pumping power requirements. The outcomes provide physics-guided design insights for achieving uniform, reliable, and energy-efficient thermal management in immersion-cooled server platforms.

2. Server board layout and components

As illustrated in Fig. 1a, the system analyzed in this study is based on the Intel® Compute Module HNS2600BPB, which integrates the Intel® Server Board S2600BPB and is housed within a 1 U half-width form factor, designed for high-density DC environments [32]. The server Printed Circuit Board (PCB) is equipped with dual LGA3647 sockets, supporting up to two Intel® Xeon® Scalable processors (1st or 2nd generation) with a maximum TDP of 165 W per CPU (Central Processing Unit). The memory subsystem includes 16 DIMM (Dual In-line Memory Module) slots (8 per processor), supporting DDR4 RDIMM or LRDIMM modules at speeds up to 2933 MT/s. The board also supports Intel® Optane™ DC persistent memory modules at speeds up to 2666 MT/s, provided the system is configured with compatible 2nd Gen Xeon processors. For platform control and I/O operations, the system incorporates the Intel® C622 chipset PCH. Onboard storage options include one M.2 slot for SATA/PCIe x4 devices, and optional bridge boards provide connectivity for additional SATA/SAS/PCIe NVMe drives, depending on the configuration. Power is supplied through the Intel® Server Chassis H2204XXLRE, which accommodates up to four compute modules and includes two 2130 W Common Redundant Power Supply (CRPS) units rated at 80 PLUS Platinum efficiency. The VR is responsible for power delivery to the CPU and memory modules. The chassis dimensions are approximately 771 mm × 86.9 mm × 438 mm. This physical architecture serves as the basis for the numerical modeling and thermal analysis presented in the following sections, with emphasis placed on chip-level thermal behavior surrounding the CPUs and DIMMs, where most heat generation occurs. Fig. 1b presents a diagram of the SPIC system layout, with the server chassis omitted for clarity.

In SPIC systems, dielectric liquids with high boiling points are typically selected to ensure the coolant remains in the liquid phase throughout operation and avoids phase change. In this work, FC-40 is utilized as the coolant due to its excellent thermal and chemical stability. FC-40 is a fully fluorinated liquid with a boiling point of approximately 165 °C, making it suitable for applications with elevated operating temperatures. Although its thermal conductivity is relatively low, its low viscosity promotes effective convective heat transfer, offering performance advantages over many conventional dielectric coolants. To account for temperature-dependent variations in FC-40's thermophysical properties, the correlations listed below are adopted [25]:

$$\lambda = 6.5458 \times 10^{-2} - (6.3265 \times 10^{-5}T) + (1.7444 \times 10^{-8}T^2) \quad (1)$$

$$cp = 1013.9205 + (1.9935T) - (3.1064 \times 10^{-3}T^2) \quad (2)$$

$$\rho = 1897.4313 - (2.1733T) + (2.5098 \times 10^{-4}T^2) \quad (3)$$

$$\mu = 7.4417 \times 10^{-3} - (2.0621 \times 10^{-4}T) + (2.9595 \times 10^{-6}T^2) - (2.2672 \times 10^{-8}T^3) + (7.1023 \times 10^{-11}T^4) \quad (4)$$

These correlations are developed based on the experimental data reported by Wang et al. [33], using linear regression analysis with a reported error of less than 1 %. The correlations are applicable within the temperature range of 0–100 °C, which encompasses the full operating range of the coolant considered in this study.

3. Research methodology

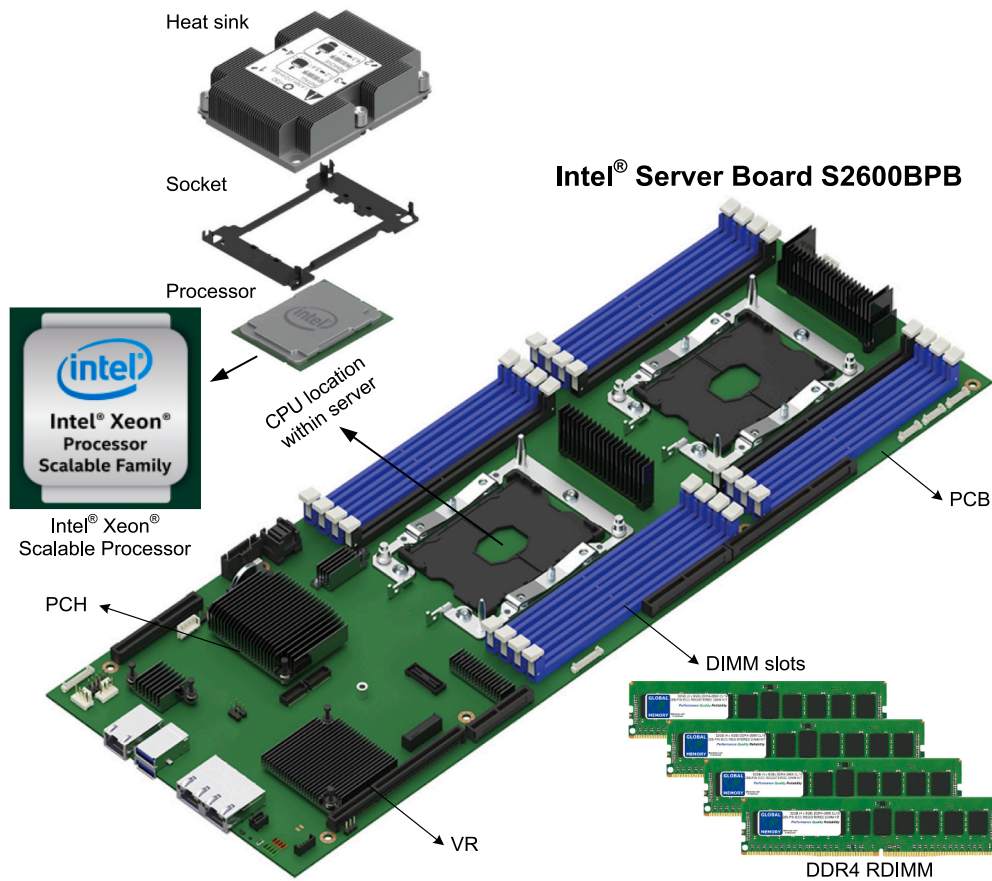
3.1. Computational model

Considering the actual operating conditions of the server examined in this study, the CPU, DIMMs, PCH, and VR are identified as the primary heat-generating components, which are cooled using heat sinks (HSs). According to component specifications and corroborating evidence from prior studies [12,14], the thermal contributions of other server components are comparatively negligible and have therefore been excluded from the computational model. Furthermore, most of these secondary components are positioned downstream of the CPUs and, due to their considerable spatial separation, exert minimal influence on the local fluid flow patterns and heat transfer behavior near the CPUs. An example of the computational model developed using ANSYS Design-Modeler, incorporating HSs with straight fins, is shown in Fig. 2. The geometric dimensions, operating powers, and material properties of the components are summarized in Table 1. The computational domain is discretized using ANSYS Meshing. In the initial mesh, the solid domain consists of structured hexahedral elements, while the fluid domain employs unstructured tetrahedral elements. Subsequently, the tetrahedral mesh in the fluid domain is converted into a polyhedral mesh using ANSYS Fluent's polyhedral meshing algorithm to enhance convergence, reduce computational cost, and maintain solution accuracy.

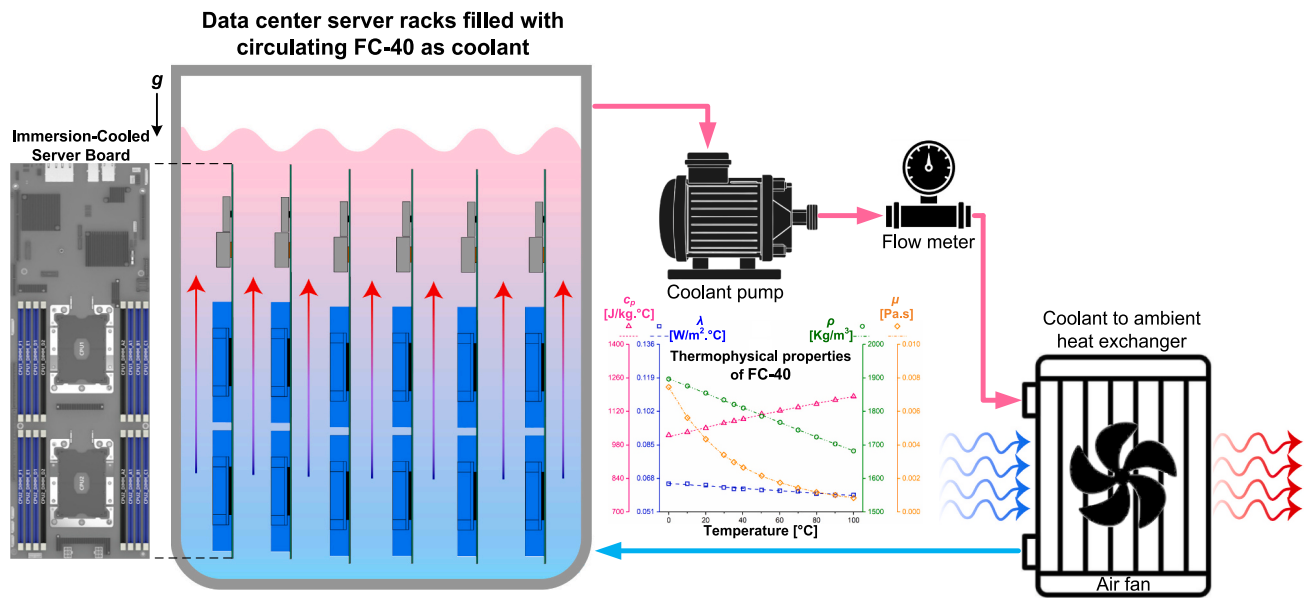
3.2. Heat sink designs and arrangements

In SPIC systems, the coolants are typically viscous, leading to significant frictional resistance as the fluid navigates through the narrow paths surrounding electronic components. Among these pathways, the HS channels attached to the CPUs present the most confined regions, creating a major bottleneck for fluid movement. Consequently, the geometric design of these HSs plays a critical role in modulating local fluid dynamics and enhancing thermal performance [34]. Beyond influencing the thermal management of CPUs, alterations to heat sink geometry can also impact the cooling efficiency of adjacent components such as DIMMs, PCH, and VR. This study investigates the application of heat sinks with oblique fins proposed for immersion-cooled server environments. The angled fin configuration induces localized converging-diverging flow channels, which actively modify the development of thermal boundary layers and turbulent structures [35]. These changes can potentially enhance or inhibit heat transfer within the system. The current research aims to investigate this design concept in detail, providing an accurate and quantitative assessment of its impact on overall server thermal performance.

Fig. 3a presents HSs featuring nineteen oblique fins designed to create converging and diverging flow channels, shown alongside a conventional heat sink with straight fins. Based on these configurations, five distinct HS arrangements within the server, illustrated in Fig. 3b, are proposed and systematically analyzed. In configuration A0, which serves as the reference case, both CPUs are equipped with HSs featuring straight channels. Configurations A1 and A2 employ HSs with oblique fins that generate converging and diverging flow paths, respectively. Configurations A3 and A4 represent hybrid setups, where converging and diverging oblique-fin HSs are combined to evaluate their integrated thermal performance.



(a)



(b)

Fig. 1. (a) Schematic layout of the server board and its components [32], and (b) diagram of the single-phase immersion cooling (SPIC) system layout.

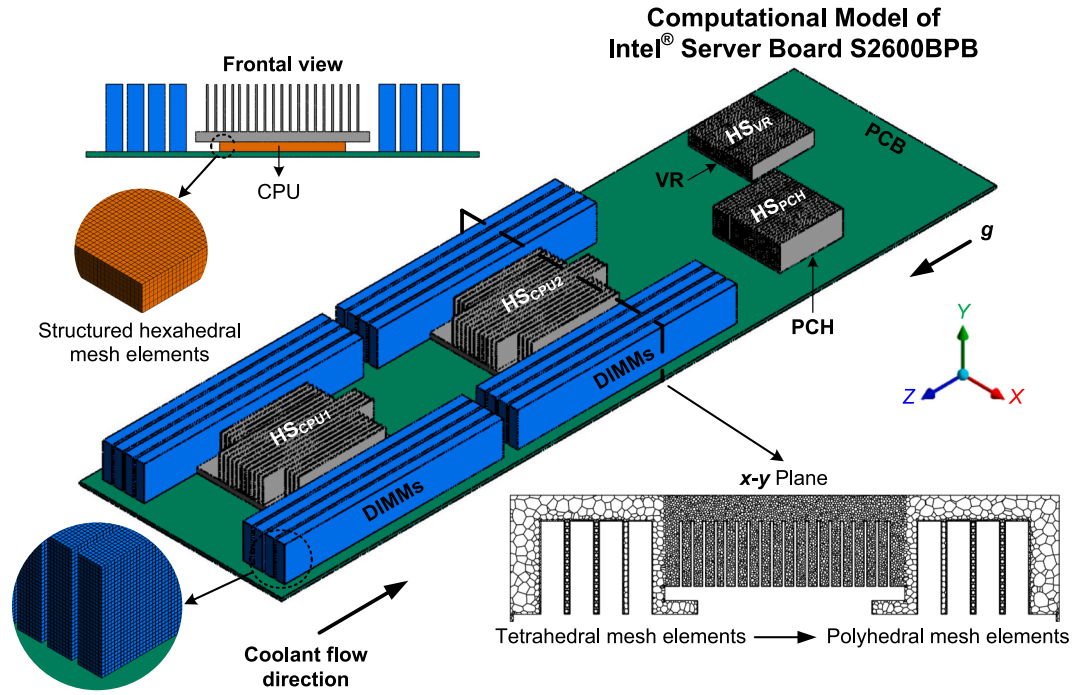


Fig. 2. Computational model and its corresponding mesh configuration.

Table 1
Details of the sever components.

Component	PCB	CPU	HS _{CPU}	DIMM	PCH	HS _{PCH}	VR	HS _{VR}
Size [mm]	625 × 176.5 × 2.4	76 × 56.5 × 4.4	108 × 78 × 4.5	191.5 × 7.5 × 30	25 × 25 × 2	55 × 58 × 22	10 × 10 × 2	55 × 55 × 12
Power [W]	0	165	0	15	10	0	4	0
Density [kg/m ³]	1850	2330	8978	2330	2330	8978	2330	8978
Thermal conductivity [W/(m·K)]	0.3	150	387.6	150	150	387.6	150	387.6
Specific heat capacity [J/(kg·K)]	1000	710	381	710	710	381	710	381

3.3. Numerical method

To assess the performance of SPIC in a server equipped with the proposed HSs, a three-dimensional, steady-state numerical simulation of coolant flow and heat transfer is developed. The model is constructed based on the following assumptions [36,37]:

- 1) The coolant flow within the computational domain is assumed to be an incompressible Newtonian fluid.

Accordingly, the governing equation for mass conservation is given as follows:

$$\nabla \cdot \vec{u} = 0 \quad (5)$$

where u is the coolant velocity.

- 2) Thermal radiation effects are neglected.
- 3) Heat losses through the side walls are assumed to be negligible.
- 4) Given the low Brinkman number, viscous dissipation effects are disregarded [23].
- 5) Heat-generating elements are treated as heat sources with uniform heat generation.

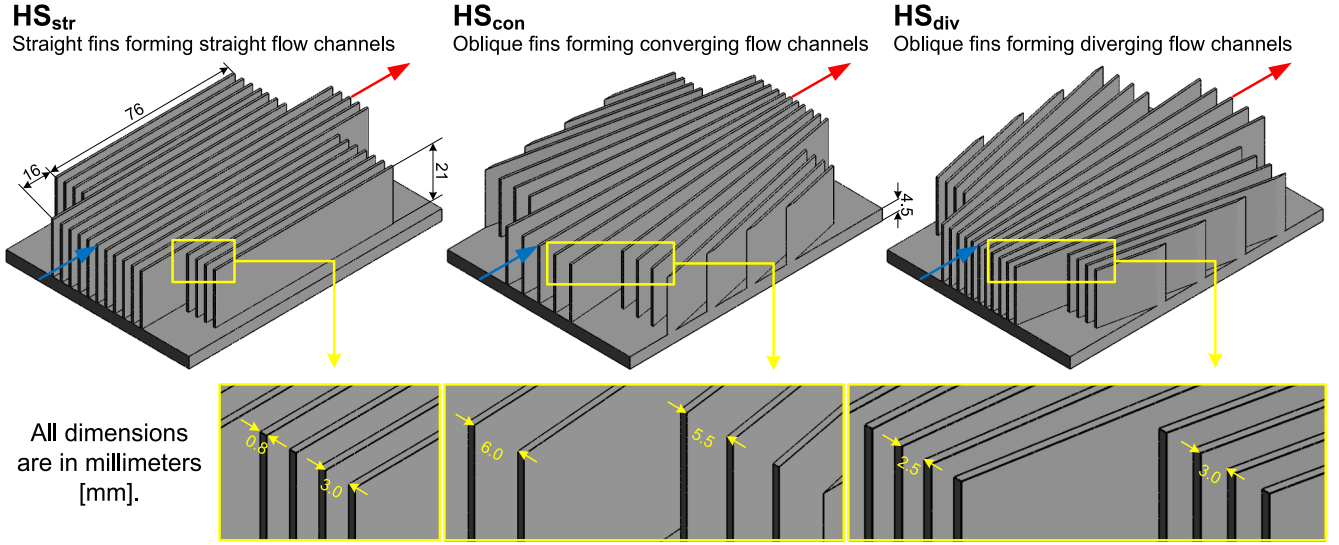
Given these assumptions, the governing equation for energy conservation is written as:

$$\nabla \cdot (\rho c_p \vec{u} T) = \nabla \cdot (\lambda \nabla T) + S \quad (6)$$

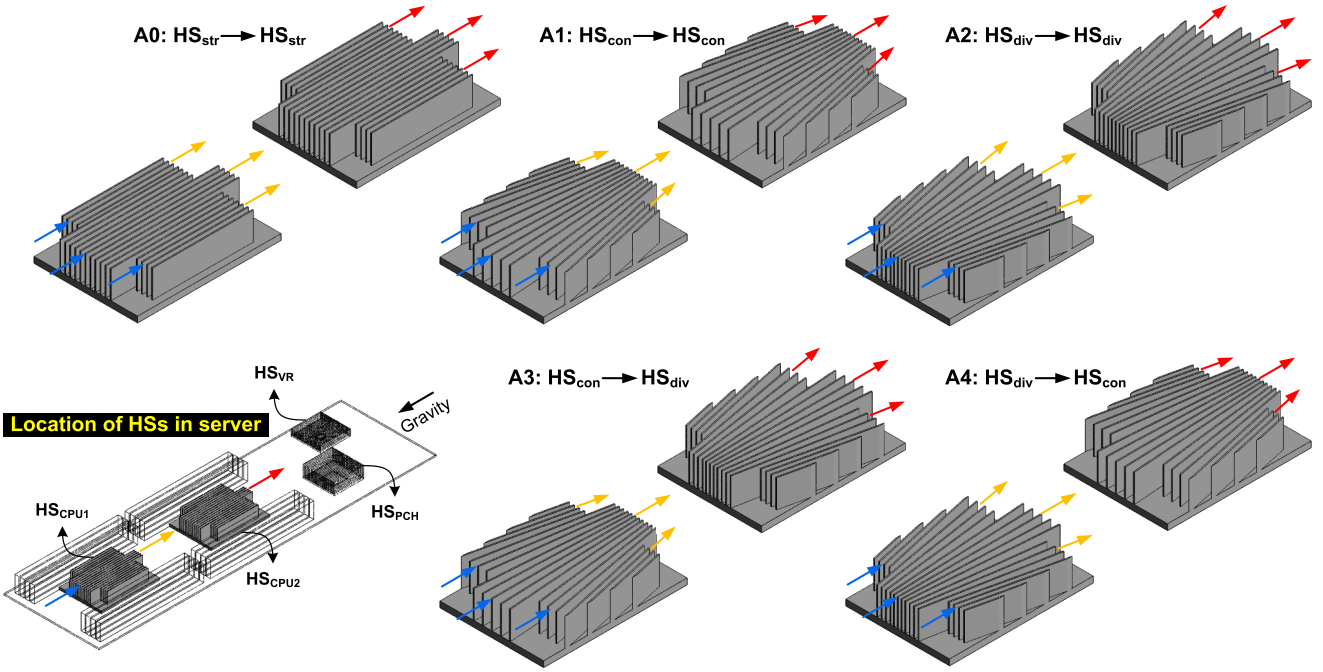
where ρ , c_p , and λ denote the density, specific heat capacity, and thermal conductivity of the fluid and solid domains, respectively; T denotes the temperature, and S is the volumetric heat source.

- 6) Gravity is oriented opposite to the direction of the coolant flow.
- 7) The complex arrangement of electronic components within the server promotes flow disturbances; therefore, turbulent flow is assumed in the analysis.

The Renormalization Group (RNG) $k-\epsilon$ turbulence model is adopted in this study due to its proven balance of computational efficiency and accuracy in predicting flow behavior in confined thermal systems [38]. Although immersion-cooled servers operate at relatively low bulk Reynolds numbers, the presence of complex geometrical features, such as heat sink fins, DIMMs, VR, and PCH, induces localized turbulent mixing, as also reported in previous SPIC studies [39]. Accordingly, the flow is modeled as turbulent using the Reynolds-Averaged Navier–Stokes (RANS) formulation. The enhanced wall treatment is applied to accurately resolve near-wall heat transfer, and the mesh is refined to achieve an average $Y^+ \approx 0.54$, ensuring proper resolution of the viscous sublayer. The governing momentum equation in the RANS framework is expressed as follows [40]:



(a)



(b)

Fig. 3. (a) Heat sink designs, and (b) their arrangements within the server.

$$\rho_f \vec{u}_j \frac{\partial \vec{u}_i}{\partial x_j} = -\frac{\partial}{\partial x_i} \left(p + \frac{2}{3} \rho_f k \right) + \frac{\partial}{\partial x_j} \left[\mu_{eff} \left(\frac{\partial \vec{u}_i}{\partial x_j} + \frac{\partial \vec{u}_j}{\partial x_i} \right) \right] + \rho_f g_i \quad (7)$$

where p signifies the pressure, and g_i represents the gravitational acceleration. k and ε denote the turbulent kinetic energy and its dissipation rate, respectively, and are expressed as:

$$\frac{\partial}{\partial x_i} \left(\rho_f k \vec{u}_i \right) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right) + G_k - \rho_f \varepsilon \quad (8)$$

$$\frac{\partial}{\partial x_i} \left(\rho_f \varepsilon \vec{u}_i \right) = \frac{\partial}{\partial x_j} \left(\alpha_\varepsilon \mu_{eff} \frac{\partial \varepsilon}{\partial x_j} \right) + C_{1\varepsilon} G_k \frac{\varepsilon}{k} - C_{2\varepsilon} \rho_f \frac{\varepsilon^2}{k} - R_\varepsilon \quad (9)$$

where G_k signifies the generation of turbulent kinetic energy due to velocity gradients, and α_k and α_ε denote the inverse of the effective turbulent Prandtl numbers for k and ε , respectively, and R_ε is correction factor in the ε equation, defined as follows:

$$G_k = \frac{1}{2} \mu_t \left(\frac{\partial \vec{u}_i}{\partial x_j} + \frac{\partial \vec{u}_j}{\partial x_i} \right)^2 \quad (10)$$

$$\left| \frac{\alpha - 1.3929}{\alpha_0 - 1.3929} \right|^{0.6321} \left| \frac{\alpha + 2.3929}{\alpha_0 + 2.3929} \right|^{0.3679} = \frac{\mu_f}{\mu_{eff}} \quad (11)$$

where μ_t the turbulent (or eddy) viscosity, and μ_{eff} denotes the effective dynamic viscosity, calculated as follows:

$$\mu_t = \rho_f C_\mu \frac{k^2}{\epsilon} \quad (12)$$

$$d \left(\frac{\rho_f^2 k}{\sqrt{\epsilon} \mu_f} \right) = \frac{1.72 (\mu_{eff} / \mu_f)}{\sqrt{(\mu_{eff} / \mu_f)^3 - 1 + C_v}} d \mu_{eff} \quad (13)$$

$$R_\epsilon = \frac{C_\mu \rho_f \eta_f^3 (1 - \eta_f / \eta_0) \epsilon^2}{1 + \beta \eta_f^3} \frac{\epsilon^2}{k} \quad (14)$$

where η_f denotes the ratio of the turbulent time scale to the mean strain time scale. The constants appearing in the aforementioned equations are summarized in Table 2. Additionally, the boundary conditions applied to the computational model are detailed in Table 3.

In ANSYS Fluent (version 2021R2), the conservation equations for mass, momentum, and energy, along with the applied boundary conditions, are iteratively solved using a pressure-based, double-precision solver based on the finite volume method. The SIMPLE algorithm is employed to handle pressure-velocity coupling using a pressure correction method. Spatial discretization is performed using a linear scheme for pressure and a first-order upwind scheme for momentum, energy, turbulent kinetic energy, and turbulent dissipation rate equations. To enhance numerical stability and facilitate convergence, the relaxation factors are assigned as follows: 0.3 for pressure, 1.0 for density, 0.7 for momentum, 0.8 for turbulent kinetic energy, 0.8 for the turbulence dissipation rate, 1.0 for turbulent viscosity, and 1.0 for energy. A convergence criterion of a residual of 10^{-6} for all variables is set to ensure a balance between computational efficiency and solution accuracy. All simulations are performed on a workstation equipped with 56 cores and an Intel® Xeon® Platinum 8173 M CPU operating at 2.00 GHz.

4. Results and discussion

4.1. Grid independence and validation study

To evaluate the impact of mesh resolution on the simulation outcomes, a grid independence study is performed. Finer meshes yield more accurate results but increase computational cost. Conversely, coarser meshes may produce inaccurate and mesh-dependent outcomes. Thus, selecting an appropriate mesh size is essential to balance accuracy and efficiency. Seven different mesh configurations are examined. The evaluation criteria included the average temperature of the CPUs (each with a TDP of 400 W) and the pressure drop of FC-40 coolant, which enters the server cabinet at 25 °C with a flow rate of 7 L/min. As summarized in Table 4, increasing the mesh density results in changes in both parameters. However, the sensitivity analysis indicates that beyond 23 million cells (corresponding to 7,067,472 cells after polyhedral mesh generation), further increases in mesh size yield negligible variations. Specifically, the difference between the 23-million and 28-million cell meshes is less than 0.5 %, equivalent to a 0.24 °C difference in CPU temperature and a 0.05 Pa difference in coolant pressure drop. Consequently, a mesh size of 7 million cells is adopted for all simulations to

Table 2
Constants for RNG k-ε turbulence model.

Constant	$C_{1\epsilon}$	$C_{2\epsilon}$	α_0	C_μ	C_v	η_0	β
Value	1.42	1.68	1.0	0.0845	100	4.38	0.012

Table 3

Boundary conditions applied to the computational model.

Surface	Boundary condition
Server cabin inlet	Velocity inlet with specified contact temperature
Server cabin outlet	Pressure outlet
Sever cabin sidewalls	Adiabatic
Coolant-electronic component interfaces	No-slip condition with coupled thermal boundary

achieve a practical trade-off between accuracy and computational time. It is noteworthy that this grid independence assessment, initially conducted for the A0 configuration, is also validated for four additional HS arrangements (A1, A2, A3, and A4). In all cases, the variations in output parameters for mesh sizes exceeding 7 million elements remain below 0.05 %.

The computational model and numerical approach are first validated using the numerical dataset reported by Chhetri et al. [41], whose results have been previously verified in multiple studies with extensive applications [42,43]. In this work, a comprehensive comparison is performed over a range of TDP values from 65 W to 150 W, at both minimum (0.5 L/min) and maximum (2.5 L/min) coolant flow rates, and for two inlet coolant temperatures of 30 °C and 40 °C. As presented in Fig. 4a, the simulation results exhibit strong agreement with the reference data. The discrepancy in predicted CPU temperatures ranges from 0.1 % to 4.5 %, which can be mainly attributed to differences in modeling assumptions and mesh discretization strategies. Despite these minor deviations, the close correlation confirms the robustness of the present numerical approach and its suitability for predicting the thermal performance of SPIC systems.

To further assess the reliability of the numerical model, additional validation is conducted against the experimental data reported by Muneeshwaran et al. [23]. The physical model used for this validation, replicating the experimental configuration, is shown in Fig. 4b. The model is evaluated under heating loads ranging from 200 W to 600 W, using FC-40 as the coolant with a flow rate of 3 L/min and an inlet temperature of 35 °C. The numerical predictions demonstrate excellent agreement with the experimental measurements. The deviation in thermal resistance remains below 3.4 %, while the maximum error in CPU temperature prediction does not exceed 1 °C. Considering inherent experimental uncertainties and the simplifying assumptions employed in the numerical formulation, these differences fall well within an acceptable range. Therefore, the validation results confirm that the proposed numerical model is accurate and dependable for predicting the thermal behavior of a data-center server under single-phase immersion cooling conditions.

4.2. Analysis of heat sink design and arrangement

Fig. 5 illustrates the impact of HS design and arrangement on CPU temperatures and their temperature uniformity, analyzed at coolant flow rates of 0.5 and 2.0 L/min. The Temperature Uniformity Index (TUI), defined as follows, is used to quantitatively evaluate thermal uniformity [44]:

$$TUI = 1 - \frac{\Delta T_{CPU}^{max}}{T_{CPU}^{avg}} = 1 - \frac{T_{CPU}^{max} - T_{CPU}^{min}}{T_{CPU}^{avg}} \quad (15)$$

where the superscripts max, min, and avg. denote the maximum, minimum, and average temperatures across each CPU, respectively. A higher TUI value (closer to 1) signifies a more uniform temperature distribution, while lower values indicate increased thermal non-uniformity.

The results demonstrate that the use of oblique-fin HSs significantly enhances cooling performance, particularly at lower coolant flow rates. For instance, at a flow rate of 0.5 L/min, all alternative configurations (A1-A4) reduce CPU temperatures by approximately 6–8 °C compared to

Table 4
Grid independent study results.

Cell number [Million]	After polyhedral mesh generation	CPU average temperatures [$^{\circ}\text{C}$]	Deviation [%]	Coolant pressure drop [Pa]	Deviation [%]
3,978,771	2,902,750	56.95	8.78	18.592	3.98
8,152,212	3,962,202	56.30	7.55	18.154	1.52
10,144,752	4,367,704	55.64	6.29	18.150	1.50
13,455,147	5,104,267	54.30	3.72	18.070	1.05
18,869,391	6,348,686	53.12	1.47	18.003	0.68
23,127,767	7,067,472	52.59	0.46	17.936	0.31
28,980,621	8,186,014	52.35	Reference	17.881	Reference

the reference case (A0). The most pronounced temperature reductions are observed for A2, with temperature decreases of 6.78°C for CPU1 and 7.62°C for CPU2. However, as the coolant flow rate increases to 2.0 L/min , the effectiveness of this configuration diminishes, with temperature reductions limited to 1.67°C for CPU1 and 2.31°C for CPU2. This trend is attributed to the dominance of forced convection at higher flow rates, where the stronger bulk fluid motion improves heat removal across all heat sinks, thereby minimizing the relative contribution of geometric enhancements such as fin inclination. In other words, as flow rate increases, the cooling benefit becomes increasingly governed by volumetric heat removal rather than localized flow manipulation, reducing the performance gap between designs.

Temperature contours in Fig. 5 clearly indicate that the maximum temperature difference within each CPU is also reduced through the use of oblique-fin HSs, consistent with the observed improvements in the TUI. For instance, at a flow rate of 0.5 L/min , the maximum temperature difference for CPU1 decreases from 9.14°C in A0 to 8.12°C in A2, while for CPU2 it drops from 8.95°C to 7.66°C . Notably, temperature uniformity enhancement becomes markedly more pronounced at higher flow rates. At 0.5 L/min , the highest TUI improvements are observed in A4 and A2 for CPU1 and CPU2, with respective increases of 0.61% and 0.70% . However, when the flow rate increases to 2.0 L/min , the improvements become significantly greater, with A1 and A4 achieving TUI enhancements of 7.71% and 6.40% for CPU1 and CPU2, respectively. This trend highlights that the thermal benefits of oblique-fin designs are flow-dependent and intensify as coolant inertia increases. At higher flow rates, the converging-channel configurations in A1 and A4 amplify velocity gradients and strengthen convective mixing, enabling more effective mitigation of downstream thermal accumulation. In other words, the enhanced flow momentum at elevated coolant speeds magnifies the geometric-induced acceleration and secondary flow structures, allowing these designs to redistribute thermal energy more efficiently and maintain a more uniform temperature field across the CPUs. Consequently, while oblique-fin heat sinks improve thermal uniformity at low flow rates, their cooling advantage becomes substantially more dominant under high-flow operating conditions typically associated with high-TDP server loads.

To evaluate the influence of HS design and arrangement on CPU temperature distribution and thermal uniformity, the coolant velocity contours on the x - z plane at $y = 20.4\text{ mm}$ are presented in Fig. 6. The outcomes clearly demonstrate that the oblique-fin HSs promote a more uniform velocity field around the CPUs. This enhanced uniformity arises from the convergent-divergent flow paths induced by the angled fin orientation, which mitigate sharp velocity gradients and improve coolant contact with the heated surfaces. These favorable flow dynamics are reflected in the temperature contours on the y - z plane at $x = 89.25\text{ mm}$, where CPU2 consistently exhibits reduced peak temperatures in arrangements featuring oblique fins. Interestingly, the DIMMs in the reference case (A0) appear to benefit from localized increases in coolant velocity in the narrow region between the HSs and the DIMM modules. This localized flow enhancement results in a temperature reduction of up to 10°C compared to arrangements featuring oblique fins, where the redirected flow predominantly targets the CPUs, leading to diminished cooling efficiency in the DIMM regions.

The influence of HS design and arrangement on peripheral components such as PCH and VR is relatively minor. However, the A1 demonstrates superior overall thermal performance, recording the lowest PCH and VR temperatures, 67.41°C and 62.28°C , respectively, under a TDP of 165 W . Additionally, A1 achieves the most effective cooling for the DIMM modules, with upstream modules (adjacent to CPU1) averaging 44.99°C and downstream modules (near CPU2) averaging 61.43°C . These observations suggest that the A1 facilitates a more optimal fluid distribution, enhancing heat removal not only from the CPUs but also from secondary components across the server. In addition, the temperature contours on the x - y planes (Panel A and Plane B) offer valuable insight into the internal thermal behavior of the HSs. In the designs with oblique fins, particularly A1, the minimal temperature gradient between the fins base and tip suggests effective heat conduction through the solid structure and efficient convective heat transfer to the coolant. A more detailed view of the coolant velocity and temperature contour variations along the server length is presented in Animation 1 (available in the online version of this article).

Animation 1. Variation of coolant velocity and temperature along the server length for different heat sink arrangements at a flow rate of 0.5 L/min under a TDP of 165 W .

To further clarify the thermal-hydraulic mechanisms across the heat sink layouts, Fig. 7 illustrates the coolant velocity variations at three transverse sections ($z = 74.25\text{ mm}$, 108.25 mm , and 142.25 mm). In the straight-fin reference (A0), the channels maintain a constant cross-section. Consequently, the velocity field remains nearly uniform along the flow direction, with only mild acceleration near the lateral edges due to reduced hydraulic resistance. As shown for HS_{str} , this uniform velocity field results in a gradual rise in convective heat transfer coefficient from the inlet toward the location where the first fin array ends. At that point, the sudden geometric interruption induces a sharp local velocity spike and the formation of complex flow structures, which in turn causes a marked increase in the heat transfer coefficient (corresponding to Fig. 8). After this peak, the coefficient briefly relaxes as the flow re-develops and rises again near the outlet where the next major disturbance occurs. However, owing to its constant geometry, A0 cannot dynamically respond to spatially varying thermal loads, limiting its capability to preserve temperature uniformity under high-TDP conditions.

In contrast, the oblique-fin arrangement in A1 (HS_{cor}) produces gradually converging channels along the flow direction. As observed in the right-hand plots, the channel contraction continuously accelerates the coolant from the upstream to the downstream regions. This controlled increase in velocity enhances forced convection and yields a progressively rising heat transfer coefficient toward the outlet side, where thermal accumulation typically intensifies due to prolonged heat exposure. As a result, the A1 layout more effectively offsets downstream heating and promotes more uniform surface temperature distribution across the CPUs. Additionally, as the coolant temperature rises along the flow path, the velocity gain compensates for the declining thermal driving force, maintaining higher heat transfer performance. As illustrated in Fig. 8, stronger downstream secondary vortices are also evident in this configuration, further boosting heat transfer in the outlet region. Conversely, the oppositely-oriented oblique fins in A2 (HS_{div}) create a

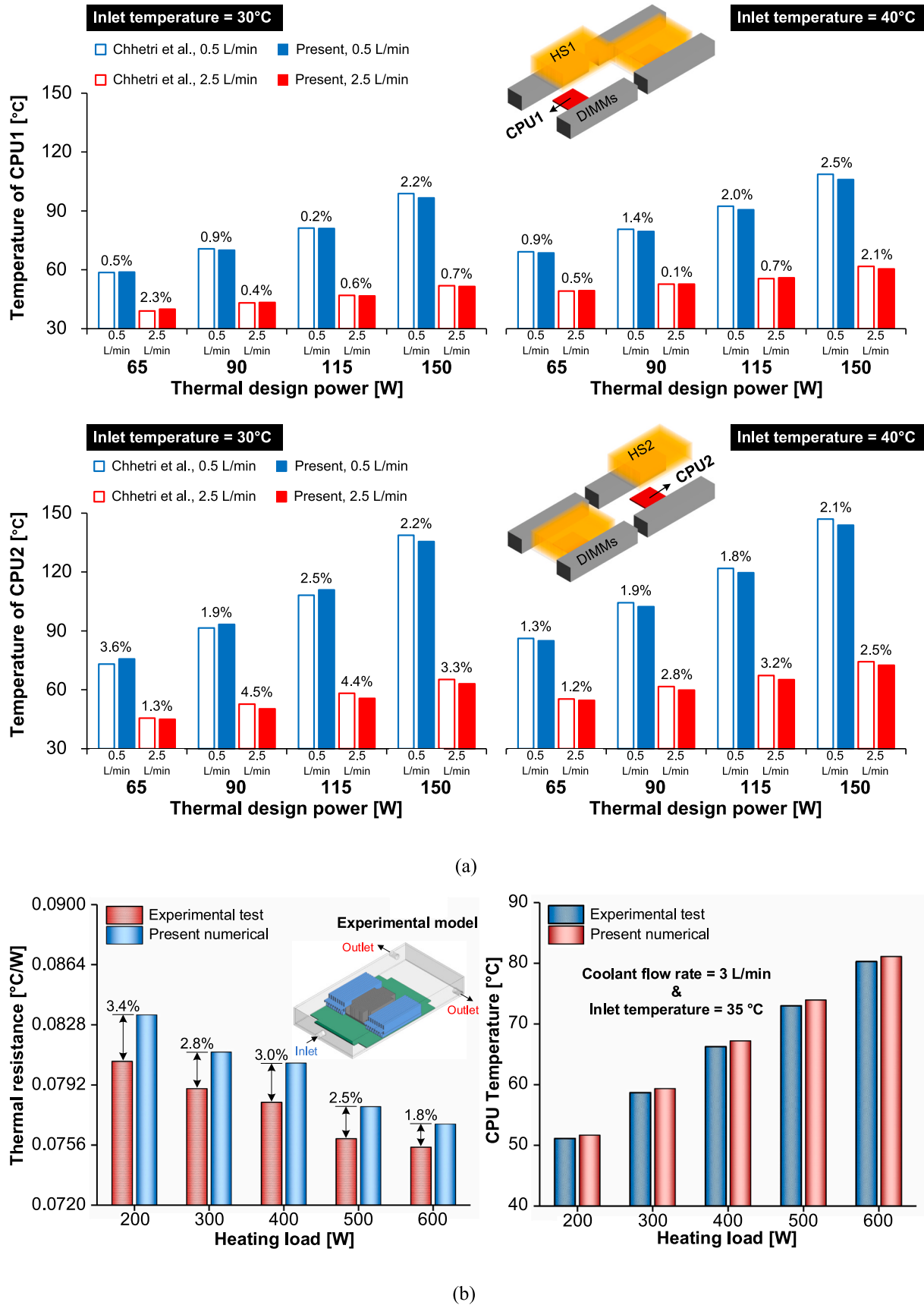


Fig. 4. Validation of the numerical model through comparison with (a) numerical reference data reported by Chhetri et al. [41] and (b) experimental reference data reported by Muneeshwaran et al. [23].

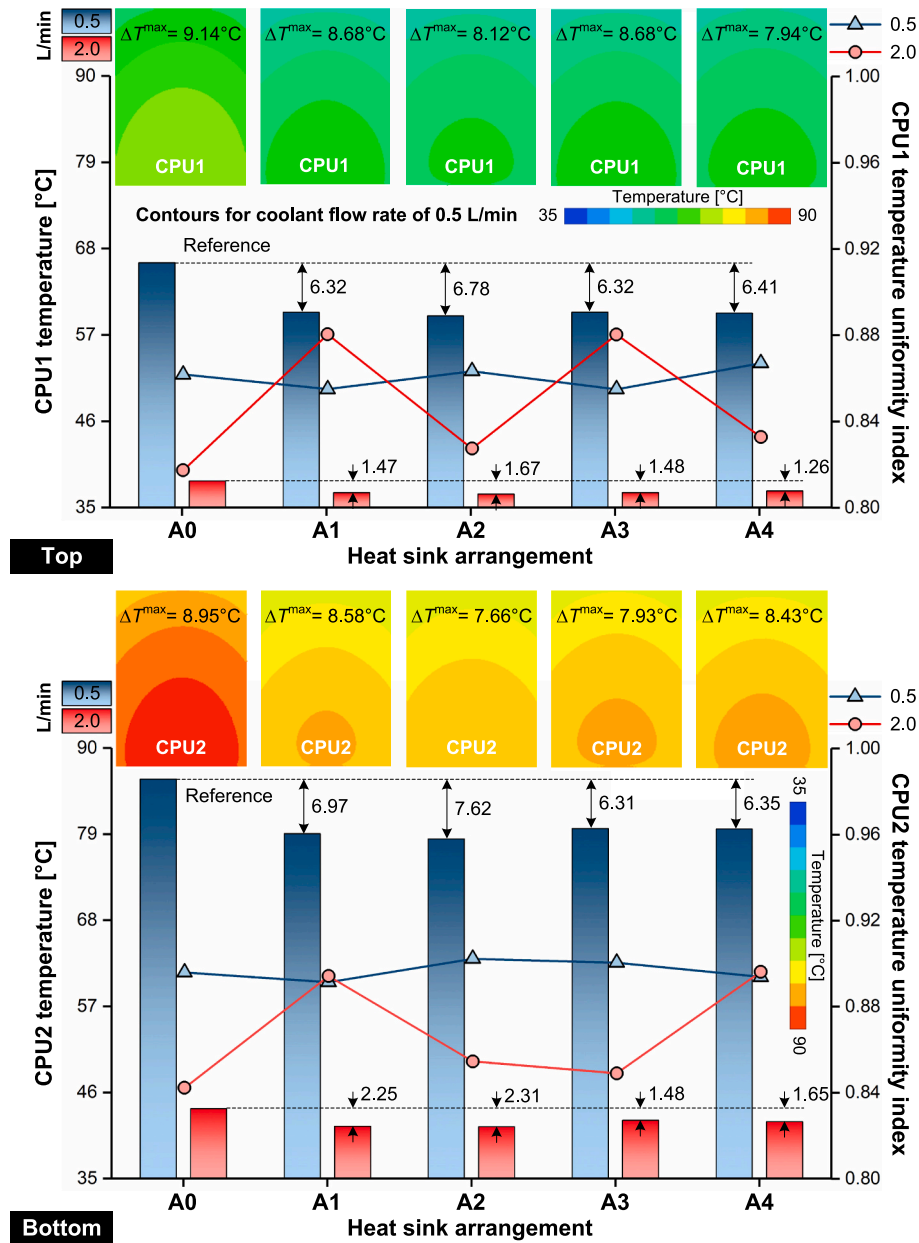


Fig. 5. Influence of heat sink design and arrangement on CPU temperatures and the corresponding temperature uniformity index under a TDP of 165 W: (Top) CPU1 and (Bottom) CPU2.

divergent flow channel. This gradual expansion reduces the coolant velocity along the axial direction, resulting in peak heat transfer near the inlet, where the flow remains relatively fast and retains swirling secondary motions. While this configuration effectively suppresses initial CPU heating by intensifying upstream cooling, the diminishing velocity downstream weakens convective transport. As reflected in the temperature contours, this can result in heat buildup near the outlet, making the A2 design less favorable for applications where downstream thermal control and uniformity are critical.

For a comprehensive assessment of thermal performance, both the average CPU temperature and the overall TUI are presented in Fig. 9. The average CPU temperature represents the mean temperature across both CPUs, while the overall TUI is calculated based on the maximum and minimum temperatures within the entire CPUs volume, offering a representative metric for evaluating thermal uniformity across the system. The results clearly demonstrate that the oblique-fin HS forming diverging channels (A2) yields the lowest average CPU temperatures,

reaching approximately 68.90 °C at a flow rate of 0.5 L/min and 39.15 °C at 2.0 L/min. In contrast, the highest thermal uniformity is observed in A1, which utilizes the oblique-fin HS forming converging channels, achieving an overall TUI of 0.76 at a flow rate of 2.0 L/min. Notably, this configuration also delivers the second-lowest average CPU temperature, following A2. This underscores their superior ability to enhance convective heat transfer and mitigate thermal hotspots. This finding suggests that while A1 may not provide the lowest absolute CPU temperatures, they are highly effective in uniformly distributing coolant across the CPU surface and minimizing localized temperature gradients.

4.3. Effects of CPUs' thermal design power

Considering the rapid advancements in technologies like AI and HPC, DC servers are facing a growing demand for CPUs with higher TDP ratings. This section evaluates the performance of the selected HSs arrangement (A1) in cooling CPUs with elevated TDP values ranging

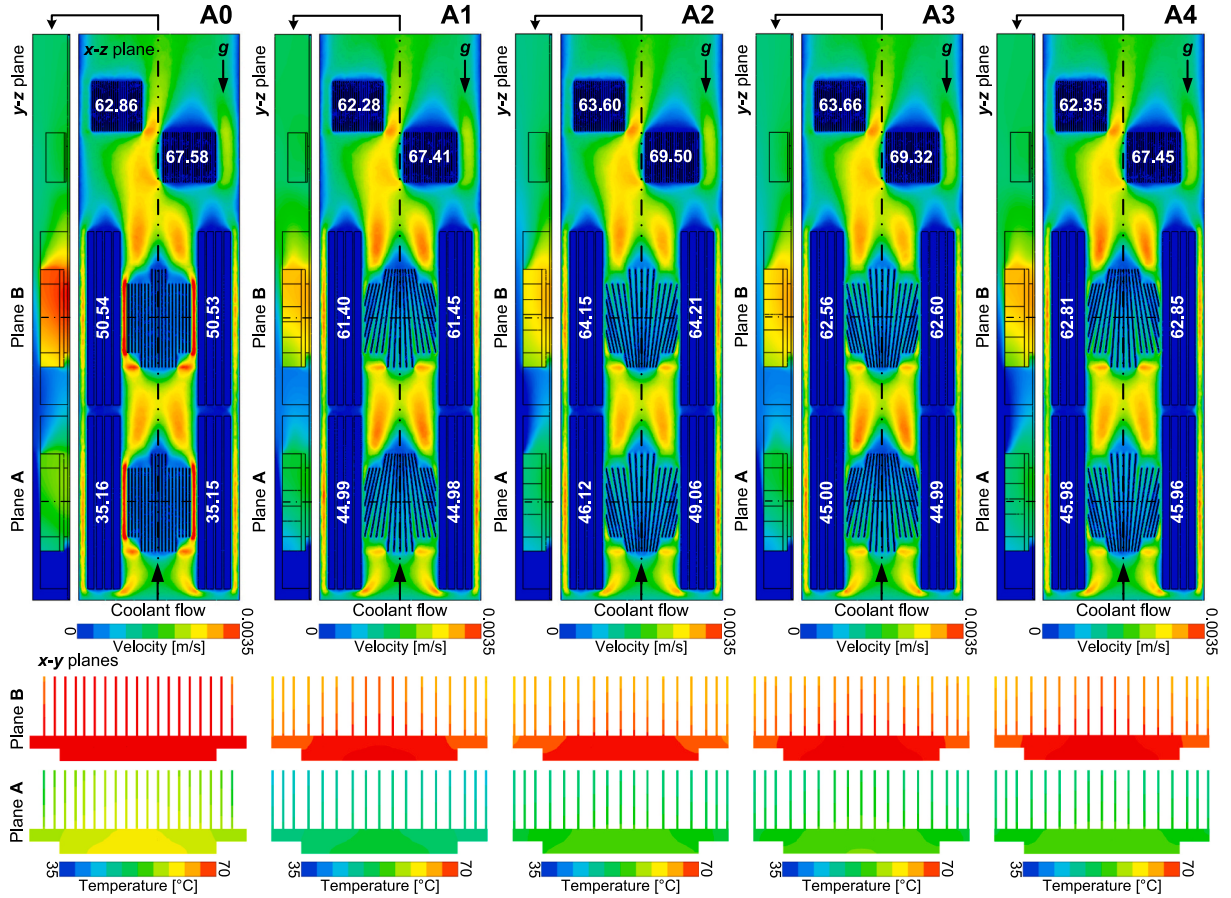


Fig. 6. Influence of heat sink design and arrangement on coolant velocity and temperature contours at a flow rate of 0.5 L/min, as well as temperature distributions of heat sinks and CPUs under a TDP of 165 W.

from 200 W to 500 W. Fig. 10 presents the results for the temperature of each CPU, their corresponding TUI, the average CPU temperature, and the overall TUI. As expected, increasing TDP results in higher CPU temperatures and lower TUI values. The findings confirm that, when using FC-40 as the coolant at an inlet temperature of 20 °C and a flow rate of 2 L/min in the A1 configuration, both CPUs can be maintained below the target temperature of 79 °C [45,46] for TDP values up to 450 W. However, at a TDP of 500 W, although CPU1 remains approximately 8.5 °C below this threshold, CPU2 exceeds it by 3.19 °C. This indicates that achieving lower CPU temperatures at higher TDPs requires an increased coolant flow rate, as discussed in the next section.

The results further show that for every 50 W increase in TDP, the temperature of CPU1 rises by approximately 5 °C, while that of CPU2 increases by about 6 °C, leading to an average CPU temperature rise of roughly 5.5 °C. The reduction in TUI due to increasing thermal load is more pronounced at lower TDP levels. Specifically, increasing TDP from 200 W to 250 W decreases the TUI of CPU1 and CPU2 by 1.64 % and 1.30 %, respectively, whereas an increase from 450 W to 500 W results in smaller reductions of 0.65 % and 0.52 %, respectively. A similar trend is observed for the overall TUI: a 50 W increase from 200 W to 250 W causes a 2.66 % reduction, while the same increase from 450 W to 500 W results in only a 1.10 % decrease.

To examine the effect of varying TDP on server thermal behavior, an internal analysis is performed on the temperature distribution of key components, including the PCB, CPUs, DIMMs, PCH, and VR. In this study, only the CPUs' thermal load is varied among the primary heat-generating components, while other component heat loads remained constant. The temperature contours in Fig. 11 clearly show that regions of the PCB with high power density in the vicinity of the CPUs exhibit markedly higher temperatures compared to other areas. This is

primarily due to the CPUs' high heat generation relative to their limited surface area, which results in localized thermal hotspots. Increasing TDP not only elevates the temperatures of the CPUs themselves but also raises the temperatures of downstream components, particularly the PCH and VR, through the thermal cascade effect from upstream heat sources [37]. In contrast, the effect on DIMM temperatures is minimal; increasing the CPUs' TDP from 200 W to 500 W results in an average DIMM temperature rise of less than 1.5 °C. Meanwhile, the PCH and VR experience notable temperature increases of approximately 10.32 °C and 9.76 °C, respectively.

The velocity vector plots on the x-y planes at $z = 210.25$ mm and $z = 435.50$ mm illustrate that secondary flow patterns are strongly influenced by their location along the server length. In the space between the two CPUs (Plane A), a pressure difference between the upstream and downstream of the inlet region induces normal swirling flows. These vortices enhance mixing between the hot fluid exiting the first HS and the cooler fluid flowing past the DIMMs, producing a more thermally uniform inlet stream for the second HS and improving CPU2 cooling. Additionally, the velocity vectors in this region show a jet-like impingement toward CPU2, which can significantly enhance the local heat transfer coefficient. Further downstream (Plane B), as the fluid passes through the dense server region, the circular secondary flows dissipate and transition into smoother streamlines. This promotes the development of thermal boundary layers near the PCH and VR, leading to increased fluid temperatures in these regions.

4.4. Effects of coolant flow rate

Given that a coolant flow rate of 2 L/min is insufficient to maintain the permissible temperature for both CPUs at a TDP exceeding 500 W,

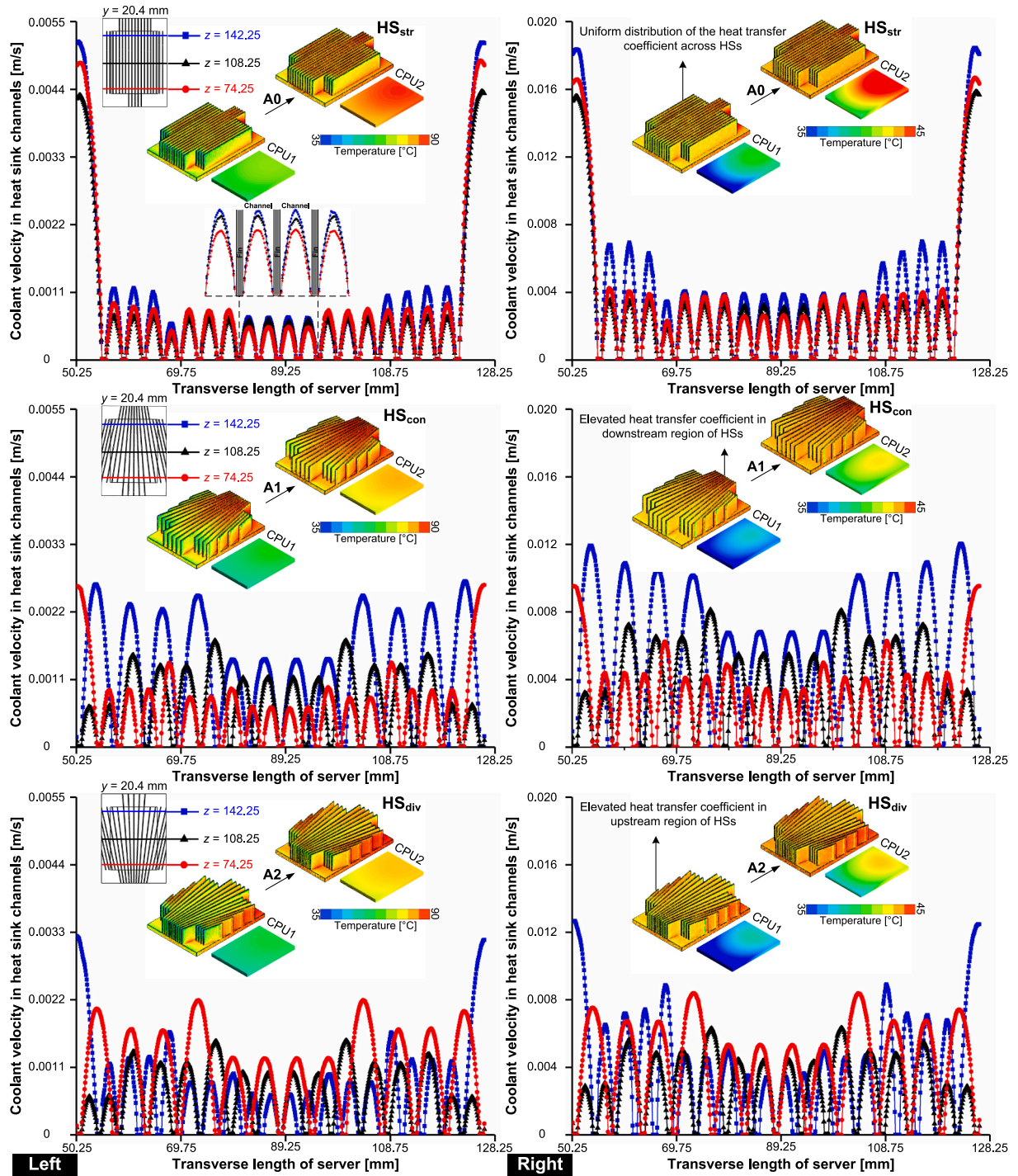


Fig. 7. Effect of heat sink design on coolant velocity within the channels, heat transfer coefficient distribution on heat sink surfaces, and CPU temperature distributions under a TDP of 165 W at coolant flow rates of 0.5 L/min (left) and 2 L/min (right).

this section examines the performance of different HS arrangements at higher flow rates. Fig. 12 presents the variations in average CPU temperature and overall TUI as the coolant flow rate increases from 2 to 7 L/min for a fixed TDP of 500 W. The results clearly show that, for all alternative arrangements (A1-A4), increasing the flow rate from 2 to 3 L/min reduces the average CPU temperature below the specified threshold (79 °C). Further increases in flow rate lead to additional reductions in average CPU temperature and improvements in the overall TUI, thereby enhancing the thermal management performance of DC servers employing the SPIC with any HS arrangement. However, the magnitude of these improvements differs across configurations. The

most pronounced effect of increased flow rate on thermal performance is observed in the reference case (A0). Specifically, increasing the flow rate from 2 to 7 L/min decreases the average CPU temperature by 36.61 % (approximately 30 °C) and improves the TUI by 17.16 %. In contrast, for alternative configurations (A1-A4), the same flow rate increase results in a temperature reduction of approximately 22 °C, with TUI improvements ranging from 14.21 % to 19.58 %. Based on these results, A0 appears most effective for achieving lower CPU temperatures at high flow rates (6–7 L/min).

However, across the entire flow rate range, the TUI of A0 remains lower than that of A1-A4, indicating the need for alternative HS

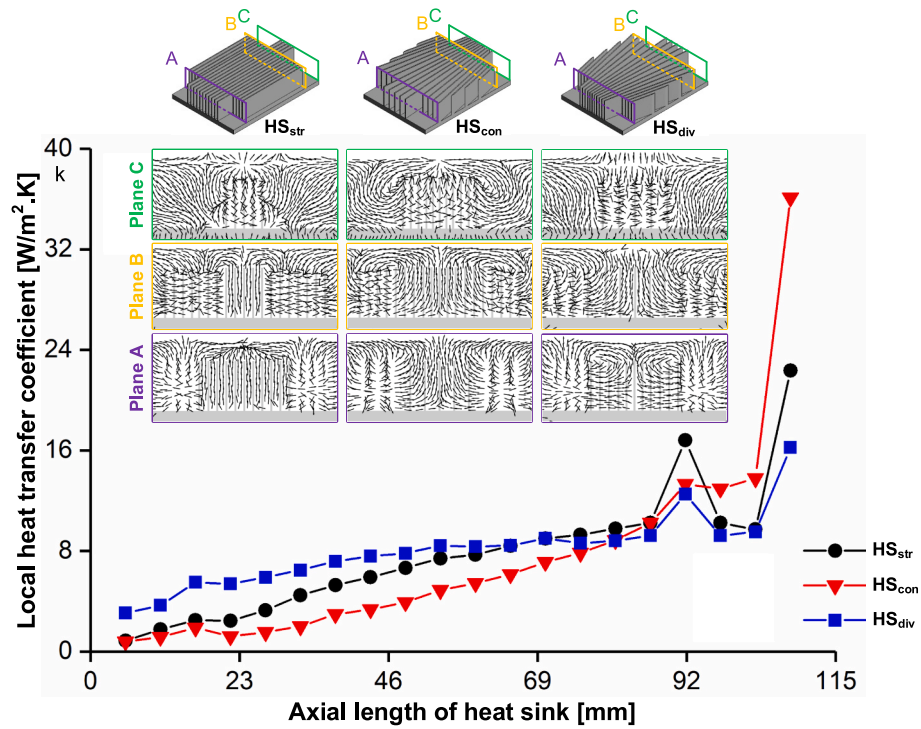


Fig. 8. Effect of heat sink design on local heat transfer coefficient along the axial length of heat sinks under a TDP of 165 W at coolant flow rates of 2 L/min.

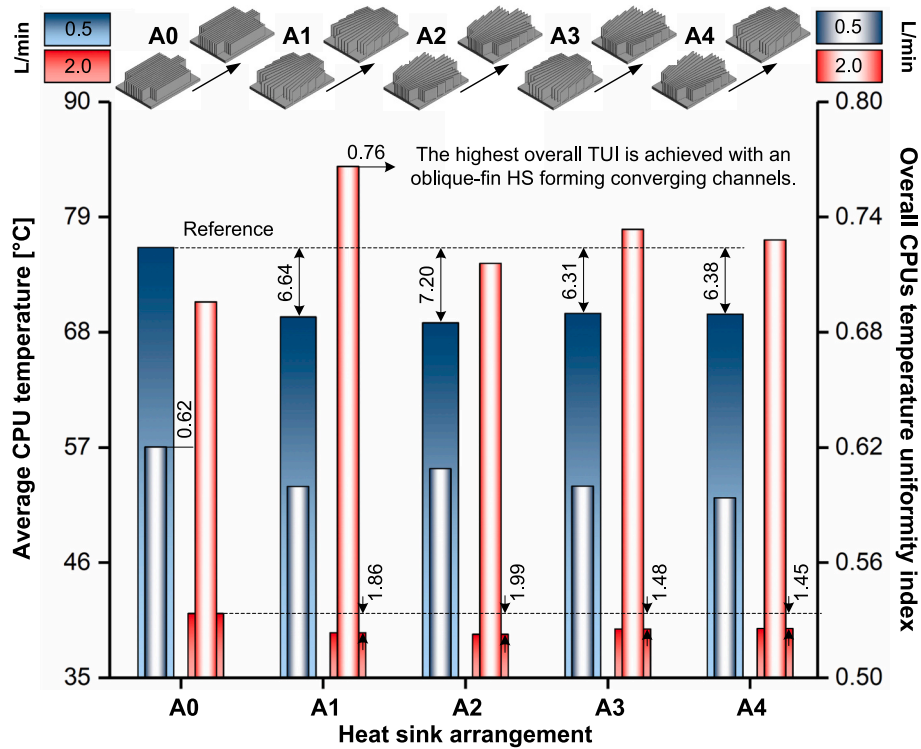


Fig. 9. Influence of heat sink design and arrangement on the average CPU temperatures and the overall CPUs temperature uniformity index under a TDP of 165 W.

arrangements to achieve better temperature uniformity. Among these, the A1 consistently provides the best TUI values. For example, at a flow rate of 2 L/min, the TUI for A1 (0.671) is approximately 2 % higher than that of A0 operating at 7 L/min (0.659). The highest TUI value is obtained with A1 at 7 L/min (0.783), which is less than 0.05 % higher than the value at 6 L/min. This indicates that further increases in flow rate

beyond 6 L/min yield negligible gains in temperature uniformity.

The pressure distribution in the three HS designs is governed by variations in the flow cross-section, which influence the coolant velocity and the associated frictional losses along the channels. In HS_{str} , the cross-sectional area remains constant, resulting in an approximately uniform flow velocity throughout the channel, as illustrated in Fig. 7.

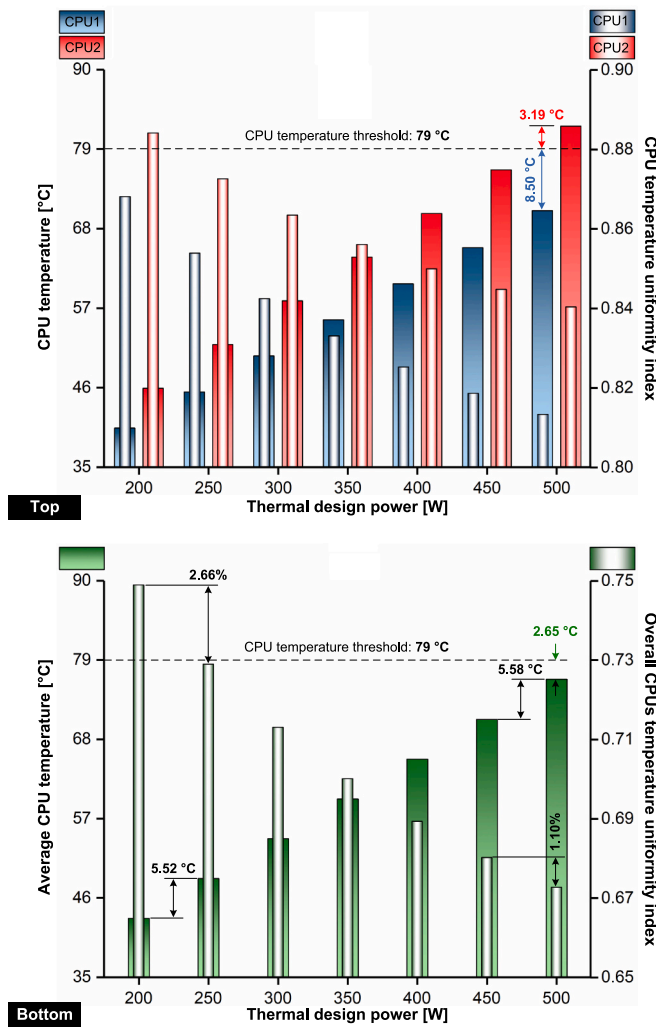


Fig. 10. Effect of TDP on CPU temperatures and the corresponding temperature uniformity index for the A1 arrangement at a coolant flow rate of 2 L/min.

Consequently, the pressure drop is primarily attributed to uniform wall friction. According to the Darcy-Weisbach relation for steady flow in constant-area channels, this produces an approximately linear pressure drop with distance. In HS_{con} , the channel cross-section gradually decreases along the flow direction, which, according to the continuity equation, increases the coolant velocity downstream. This velocity increase amplifies the dynamic pressure losses. However, in the downstream region, where the velocity is higher, the relative contribution of frictional losses is lower compared to the upstream section. This interplay results in a nonlinear pressure profile with downward concavity, characterized by a moderate initial pressure drop (37.5 % in the first half) followed by a more pronounced drop (62.5 % in the second half) toward the channel outlet. In HS_{div} , the cross-section expands along the flow direction, leading to a gradual reduction in coolant velocity. As the velocity decreases, dynamic pressure losses diminish, and frictional losses become the dominant factor in the downstream region. This results in a nonlinear pressure profile with upward concavity, characterized by a relatively steep initial pressure drop (71.5 % in the first half) followed by a reduced pressure drop rate (28.5 % in the second half) toward the outlet. The pressure distributions for all three HS designs are shown in Fig. 13, along with the corresponding pressure contours on plane A ($z = 54.5$ mm) and plane B ($z = 164.5$ mm), located at the upstream and downstream ends of the heat sinks, respectively. However, an examination of the overall pressure drop characteristics indicates that these local pressure variations do not significantly influence the

hydraulic performance of the server cabinet. As a result, the pumping power required to circulate the coolant, calculated using Eq. (16), remains nearly identical across all heat sink arrangements. The results presented in Fig. 13 quantitatively confirm this observation. The use of c A1, which exhibits the highest pumping power demand, leads to only a marginal increase in power consumption of 2.71 % and 3.01 % at flow rates of 2 L/min and 7 L/min, respectively, compared to A0.

$$P_p = V \Delta p \quad (16)$$

Finally, Fig. 14 presents the variation in temperatures of other major server components with increasing the coolant flow rate for different HS arrangements. As shown, the temperatures of these components decrease noticeably as the flow rate increases. However, the magnitude of the temperature reduction varies among components. The results indicate that, after the CPUs, the most significant temperature reductions occur in the downstream components (PCH and VR) followed by the DIMMs located upstream. The DIMMs, with a thermal load of 15 W, exhibit an average temperature reduction of only about 8.68 °C as the flow rate increases from 2 to 7 L/min. This relatively small change is due to their lower heat generation and the correspondingly smaller enhancement of convective cooling for components with lower initial temperature rise. In contrast, the downstream PCH and VR components show larger temperature drops of approximately 18.48 °C and 16.38 °C, respectively, which are greater than those of the DIMMs. This can be attributed to two factors: (i) the increased local heat transfer coefficient in their heat sinks resulting from higher coolant velocities, and (ii) the mitigation of the thermal cascade effect, whereby a higher flow rate reduces the temperature of the coolant entering their heat sinks. Both effects enhance convective heat transfer around these components at elevated flow rates.

The largest temperature reductions with increasing coolant flow rate are observed for the A2, amounting to 9.61 °C, 21.44 °C, and 17.23 °C for the DIMMs, PCH, and VR, respectively, as the flow rate increases from 2 to 7 L/min. These trends are clearly supported by the temperature contours shown in the top section of Fig. 14. Nevertheless, the A1 consistently yields the lowest temperatures for these components across all flow rates. For example, at 2 L/min, the DIMMs, PCH, and VR temperatures in A1 are reduced by 0.83 °C, 1.20 °C, and 2.81 °C, respectively, compared to A0. At 7 L/min, these differences decrease to 0.35 °C, 0.37 °C, and 1.53 °C, respectively.

4.5. Comparative study

Compared with the existing methods reported in recent literature, the proposed oblique-fin heat sink offers a fundamentally different and highly practical cooling strategy. As summarized in Table 5, previous studies have relied heavily on active enhancement techniques such as jet impingement [47], ducted propellers [48], suction fans [23], and modified flow routing [49], all of which require additional pumping power, introduce mechanical complexity, generate noise or vibration, and pose reliability risks such as fan failure or nonuniform flow distribution. Some studies have used heat sink modifications, such as circular or elliptical pin-fins [50], to improve thermal performance without external pumping. In contrast, our oblique-fin design further enhances single-phase cooling efficiency while requiring no moving components or external momentum. Although jet impingement and dual-propeller configurations achieve larger hotspot suppression (up to 38–55 % temperature improvement), these benefits come at the cost of significantly higher flow rates, increased system power consumption, and potential maintenance challenges. Two-phase designs such as circular pin-fins also show high thermal performance but suffer from vapor entrapment, instability, and limited applicability to servers operating under single-phase constraints. Within this context, the proposed oblique-fin design demonstrates a more balanced and energy-efficient trade-off, achieving meaningful temperature reductions (up to 7 °C) with less than 3 % pumping-power penalty, outperforming all active techniques in

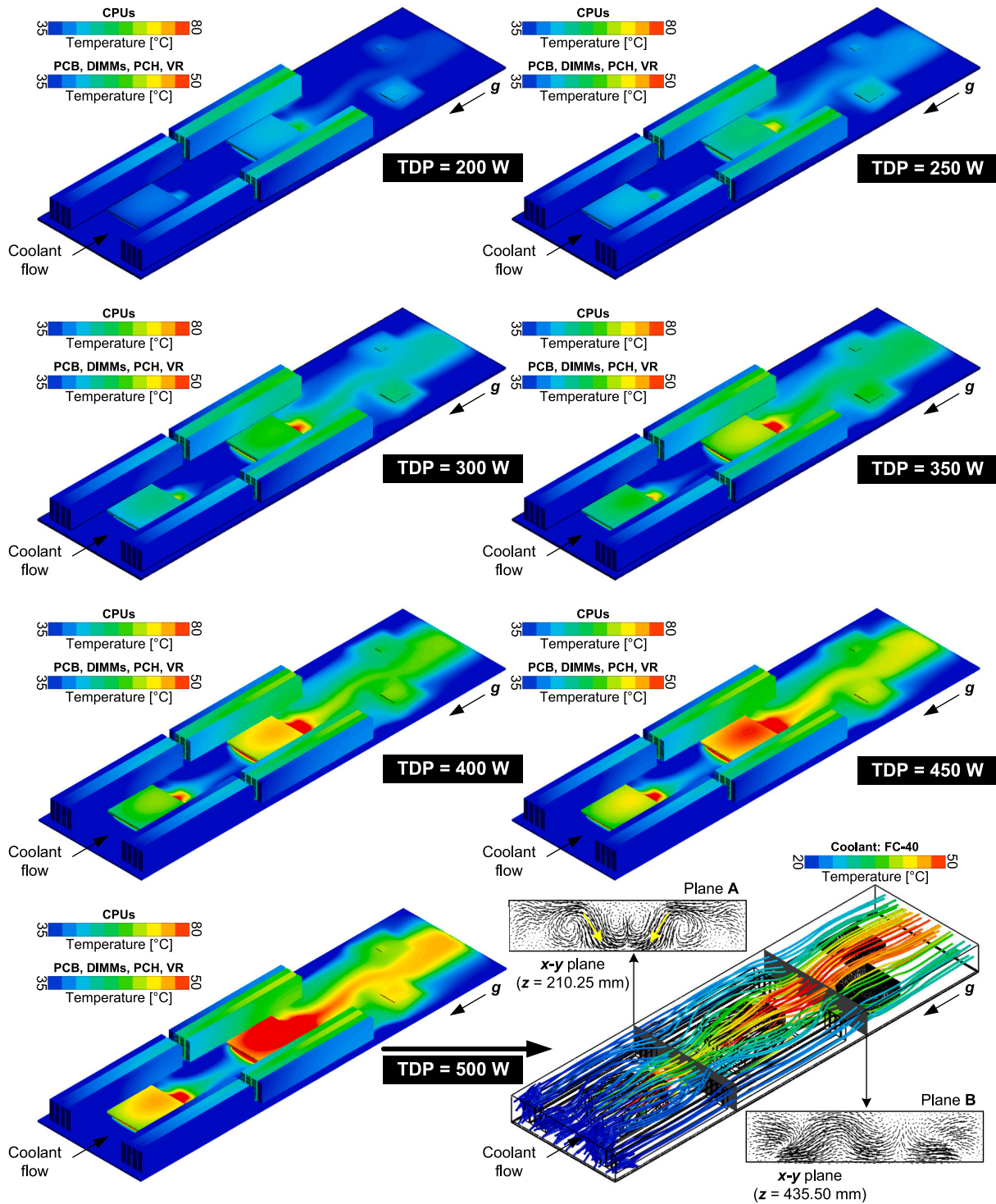


Fig. 11. Effect of TDP on the temperature distribution of the server components (CPUs, PCB, DIMMs, PCH, and VR) and coolant (FC-40) for the A1 arrangement at a coolant flow rate of 2 L/min.

hydraulic efficiency. Moreover, unlike studies that focus solely on CPU cooling, our configuration simultaneously alleviates downstream thermal cascade effects, significantly lowering the temperatures of DIMMs, PCH, and VR components without any mechanical assistance. While its absolute temperature drop is smaller than that of dual propellers or anti-gravity SPIC configurations, the proposed solution remains structurally simple, passive, low-cost, and highly reliable qualities that are essential for large-scale data-center deployment. Overall, the results confirm that the oblique-fin geometry provides a robust and manufacturable pathway

to uniform and efficient cooling, offering a strong compromise between performance gain, system simplicity, and operational stability.

5. Conclusion

This study presents a comprehensive investigation of single-phase immersion cooling (SPIC) for the thermal management of multi-CPU server platforms, focusing on the Intel® Server Board S2600BPB equipped with oblique-fin heat sinks arranged in innovative

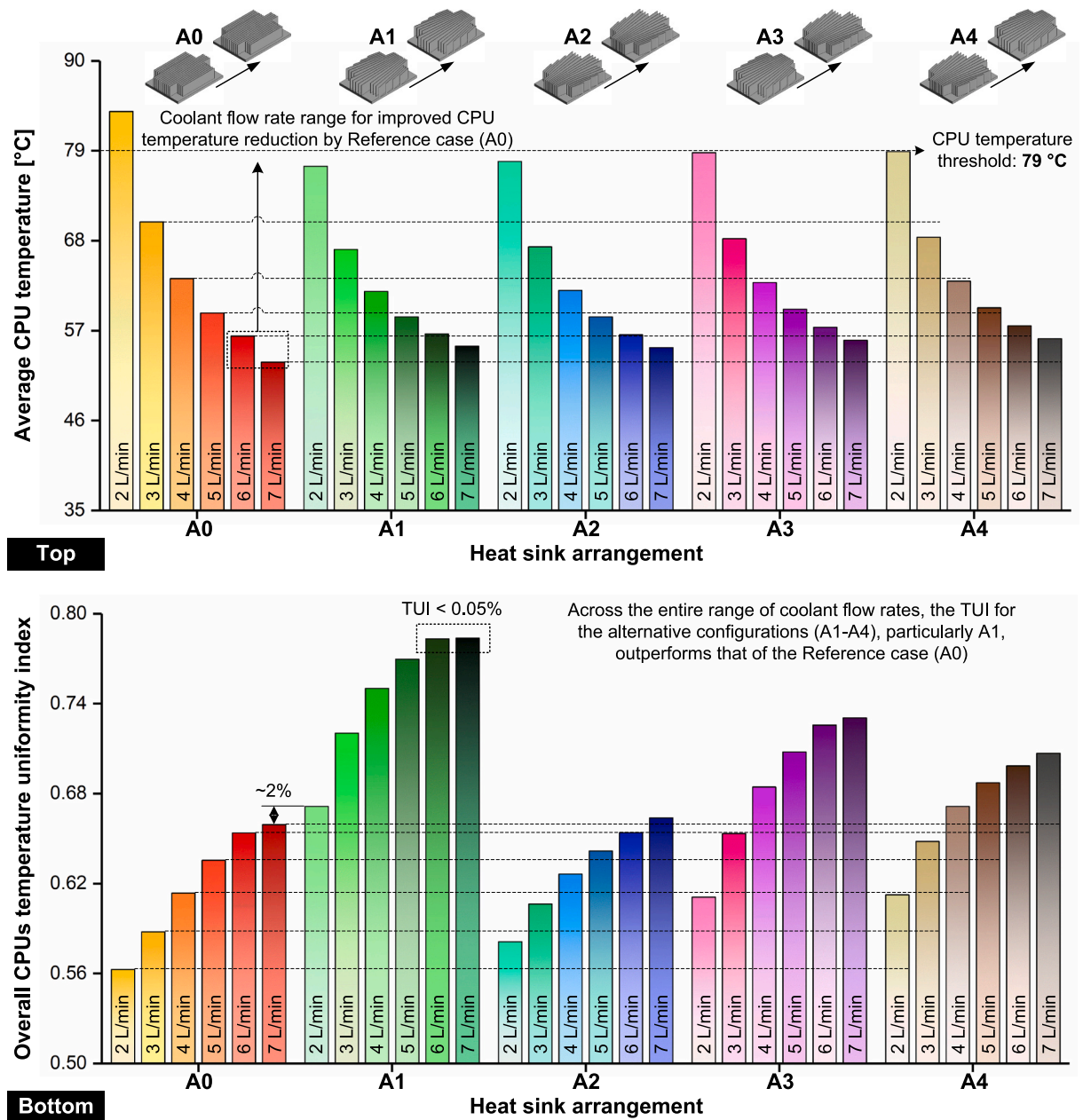


Fig. 12. Effect of coolant flow rate, heat sink design and arrangement on the average CPU temperature and the corresponding overall temperature uniformity index at a TDP of 500 W.

configurations. Using FC-40 as the coolant, the study examines the combined effects of heat sink geometry and coolant flow rate on component temperatures, the temperature uniformity index (TUI), and pumping power requirements under various CPU thermal design power (TDP) levels. The primary objectives of this research are to (i) maintain CPU peak temperatures below the specified threshold under both nominal and high-load conditions, (ii) enhance temperature uniformity relative to the baseline configuration, and (iii) minimize the increase in pumping power compared to the reference case. The key findings are summarized as follows:

- Oblique-fin heat sinks markedly improve cooling performance and temperature uniformity, particularly at low coolant flow rates, where the A2 arrangement achieves CPU temperature reductions of up to 8 °C under the nominal CPU TDP of 165 W. Although the relative temperature reduction benefit declines at higher flow rates,

temperature uniformity gains become more pronounced, with TUI improvements of up to 7.71 %.

- The A1 arrangement effectively cools CPUs with TDPs up to 450 W below the target temperature of 79 °C at a coolant flow rate of 2 L/min. However, at 500 W TDP, one CPU exceeds this limit, indicating the need for increased coolant flow at higher thermal loads. CPU temperatures rise approximately 5–6 °C per 50 W increase in TDP, while TUI declines more sharply at lower TDP levels, with diminishing reductions as TDP approaches higher values.
- Increasing CPU TDP significantly raises temperatures of downstream components such as the PCH and VR due to thermal cascade effects, with temperature increases of about 10.32 °C and 9.76 °C respectively, while DIMM temperatures remain largely unaffected. The A1 arrangement promotes optimal coolant distribution, effectively enhancing heat removal from both CPUs and secondary components within the server.

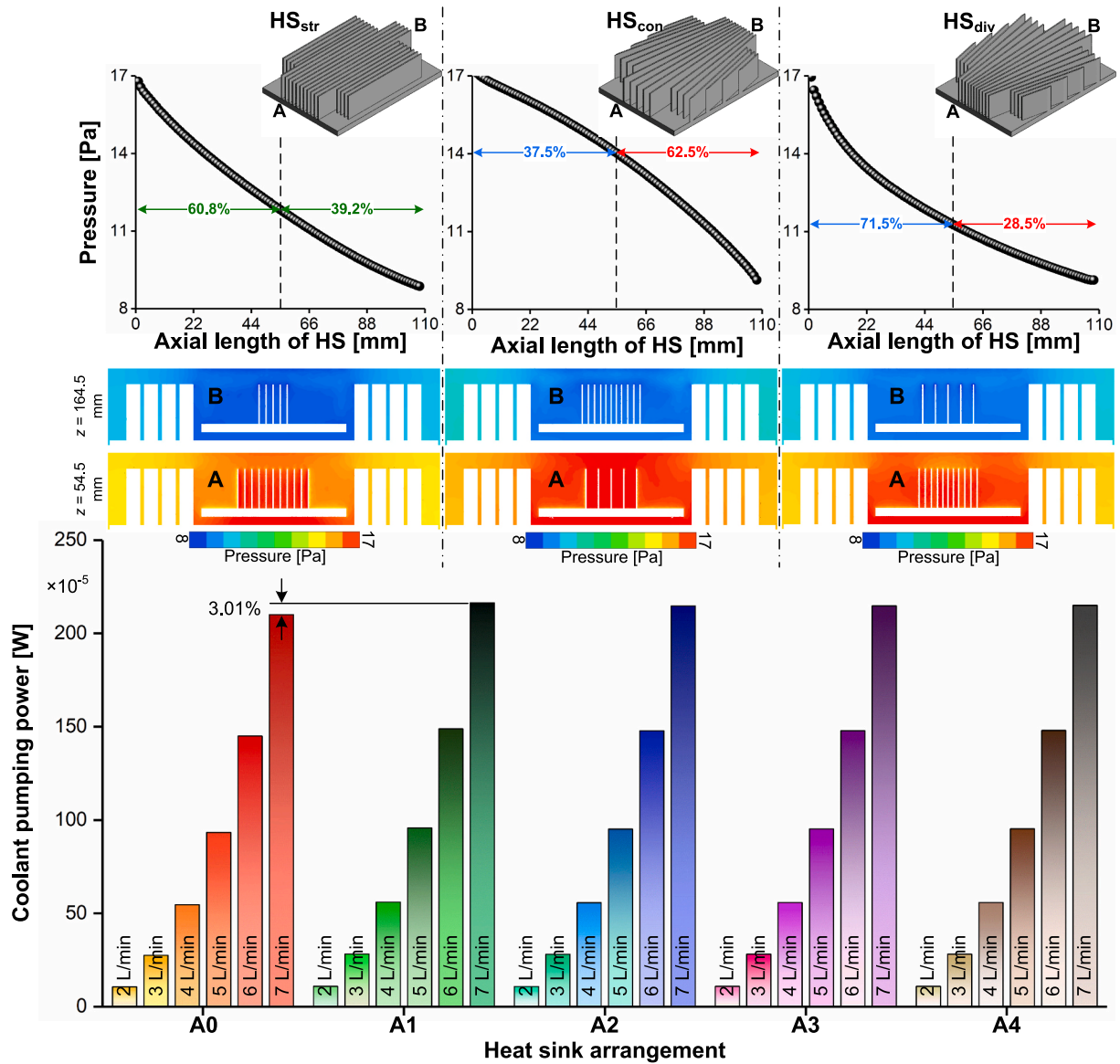


Fig. 13. Effect of coolant flow rate, heat sink design and arrangement on the coolant pressure drop and pumping power at a TDP of 500 W.

- Increasing the coolant flow rate from 2 to 3 L/min reduces average CPU temperatures below the target threshold for all heat sink arrangements, with further increases in flow rate enhancing both temperature reduction and uniformity. While the reference case achieves the largest temperature drop at high flow rates (up to approximately 30 °C reduction from 2 to 7 L/min), it consistently exhibits lower TUI values compared to the alternative arrangements. Among these, the A1 arrangement provides the best temperature uniformity across all flow rates, with TUI improvements plateauing beyond 6 L/min.
- Increasing the coolant flow rate substantially reduces the temperatures of major server components, with downstream components such as the PCH and VR exhibiting significantly greater temperature reductions compared to upstream DIMMs. At a CPU TDP of 500 W, the DIMMs experience an average temperature decrease of 8.68 °C, whereas the PCH and VR demonstrate more pronounced reductions of approximately 18.48 °C and 16.38 °C, respectively. The A1 arrangement consistently maintains the lowest absolute temperatures across all flow rates, highlighting its superior cooling effectiveness.

- Local pressure variations caused by different heat sink designs have a minimal impact on the overall hydraulic performance of the SPIC, resulting in nearly identical pumping power requirements across all configurations. Notably, the A1 arrangement, which demands the highest pumping power, increases power consumption by only 2.71 % and 3.01 % at flow rates of 2 and 7 L/min, respectively, compared to the reference case.

In conclusion, heat sinks featuring oblique fins that form converging channels not only provide effective CPU cooling with improved temperature uniformity but also enhance the thermal management of other server components. This configuration achieves significant thermal benefits while incurring only a minimal hydraulic penalty. Future studies could extend the application of these heat sinks to alternative coolants by optimizing their geometric parameters, followed by experimental validation through manufacturing and testing under real operating conditions.

CRediT authorship contribution statement

M. Khoshvaght-Aliabadi: Writing – review & editing, Visualization,

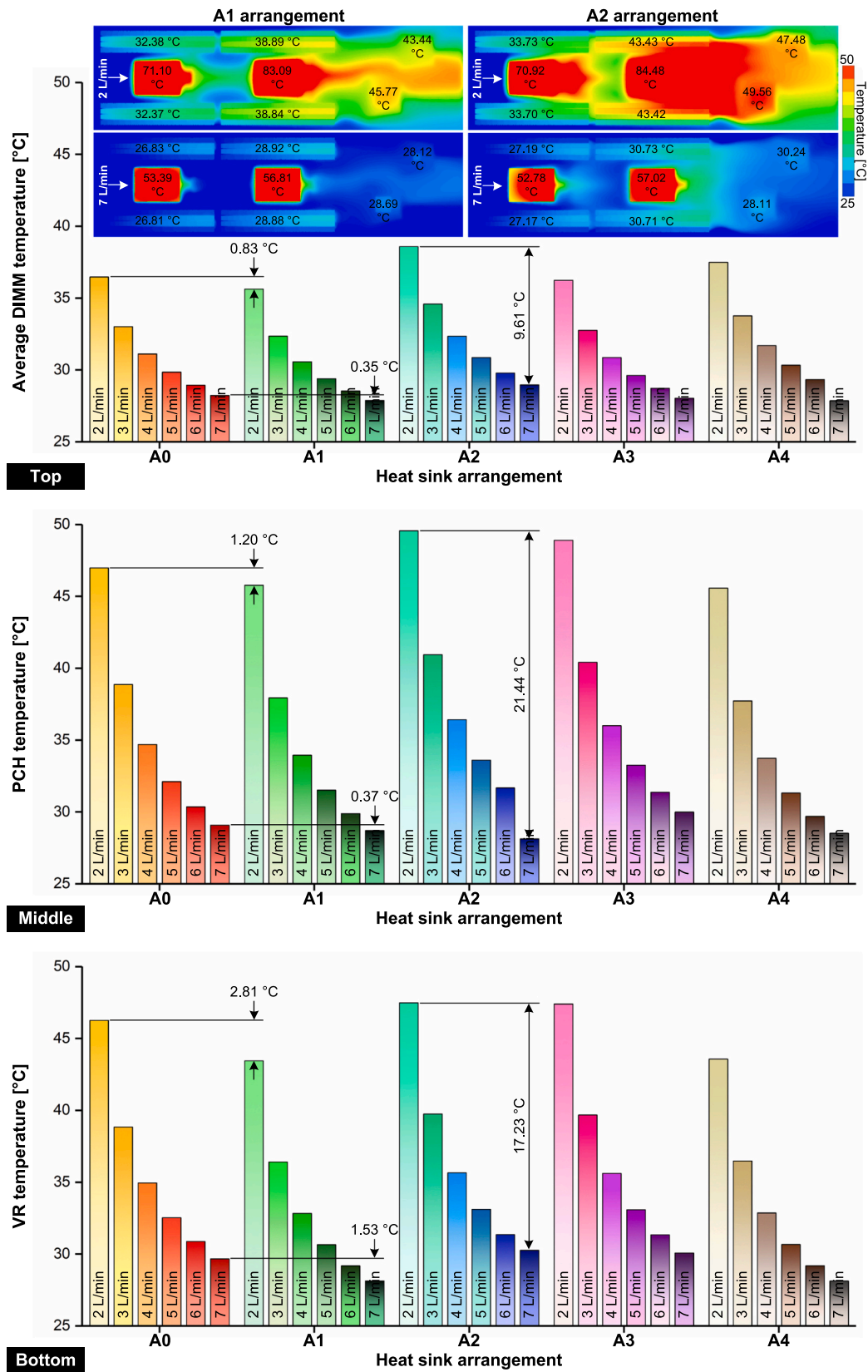


Fig. 14. Effect of coolant flow rate, heat sink design and arrangement on the average DIMM, PCH, and VR temperatures at a TDP of 500 W.

Table 5

Comparison of thermal and hydraulic performance of various CPU heat dissipation methods.

Ref.	Technique	TDP [W]	Average CPU Temperature [°C]	Pumping Power [W]
Jin et al. [47]	Jet impingement	500	Reduced by 38.74 %	Lowered by 7.81 %
Jin et al. [48]	Ducted propellers	400	For single propellers reduced to 6.2 °C and for dual propellers 15.6 °C	Maximum power consumption for Single propellers was 0.31 W, and for dual propellers was 0.56 W
Muneeshwaran et al. [23]	T- and Z-flow configurations, bypass flow control	200–600	Reduction of 0.5–2.8 °C achieved in the T-configuration, The flat plate experienced 60 °C	–
Huang et al. [49]	Opposite-flow SPIC configuration	350	Reduced the chip case temperature by 9.3 %	–
Shete et al. [50]	Vertical plain heat sink	18–45	The surface temperature Reduced by approximately 35 %	–
Present study	Oblique-fin heat sink	500	Reduced CPU temperature by 7 °C and increased TUI by 20 %	3 % increased vs. reference model

Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **A. Nasrolahzadeh:** Validation, Software, Investigation. **P. Ghodrati:** Writing – original draft, Data curation. **Y.T. Kang:** Writing – review & editing, Formal analysis.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to modify the English of paper. After using this service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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